A Wearable EMG-Morse Code Silent Communication with AES Security

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***Abstract*— Secure and reliable communication is essential in both civilian and defense contexts, yet traditional methods often face challenges such as interception risks, reliance on strong network connectivity, and limited portability. To address these limitations, this work explores an alternative communication approach based on human muscle signals. Electromyographic (EMG) activity is used to generate Morse code patterns, providing a simple and effective way to transmit information even in any constrained environments. To ensure confidentiality, the system integrates AES encryption, transforming the Morse-coded messages into secure data before transmission. By combining EMG sensing, lightweight coding, and modern cryptographic protection, the proposed system offers a portable, discreet, and robust communication method suitable for scenarios where conventional systems may fail. The approach demonstrates strong potential for both civilian applications and defense operations requiring secure and adaptable communication.**

Keywords— Electromyography(EMG), AES Encryption, Real-time Communication.

I. INTRODUCTION

Communication plays a fundamental role in both civilian society and defence operations, enabling coordination, decision-making, and information exchange across diverse environments. As the demand for secure and resilient communication continues to grow, traditional systems often reveal inherent limitations. Many existing communication methods are susceptible to interception, rely heavily on stable network connectivity, or fail to operate effectively in remote, congested, or hostile conditions. These challenges highlight the urgent need for alternative solutions that provide reliability, confidentiality, and independence from conventional infrastructure.

Advancements in human-machine interaction have created new opportunities to address these constraints. By leveraging physiological signals produced naturally by the human body, it is possible to establish communication channels that are inherently discreet, portable, and less dependent on external networks. Electromyographic (EMG) signals, generated by muscle activity, offer a compelling medium for such interaction. These signals can be processed and transformed into structured communication patterns, allowing users to transmit information using minimal physical movement.

Morse code, despite its historical origins, remains an effective and robust communication method, particularly in scenarios with limited bandwidth or degraded signal conditions. Its simplicity, adaptability, and low-complexity requirements make it suitable for integration with modern signal-processing systems. When combined with secure encoding techniques, Morse-based communication can be transformed into a highly reliable and confidential mode of information exchange.

The system proposed in this work integrates EMG-based signal acquisition with Morse code generation and secure message encryption using the Advanced Encryption Standard (AES). By converting subtle muscle movements into encrypted communication packets, the approach ensures both usability and data confidentiality. This framework bridges traditional communication principles with contemporary security practices, offering a versatile solution capable of operating in both civilian and defence environments where reliability, security, and adaptability are essential.

II. D ESIGN METHODOLOGY

The details of working system and design can be explained as following:

A. Block Diagram

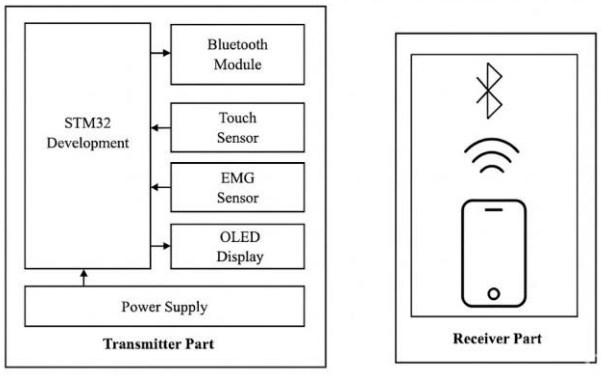

Fig 2.1. Block Diagrams of the developed device

The block diagram of the developed device are presented in Fig 1. The overall working principle of the device is when the microcontroller receives the data from the EMG sensor along with touch sensor, it processes the incoming data and encrypts it sends via Bluetooth to display it on the mobile screen.

A. Power Supply: The system is powered by a rechargeable battery or an external DC source. Since the sensors and the

microcontroller require regulated voltages, the input supply is conditioned to provide stable power levels, typically 5 V or 3.3 V, to ensure safe and reliable operation

B. Buck Converter: A buck converter is used to step down the input voltage to the appropriate levels needed by the system components. This DC-DC converter protects sensitive devices, such as the STM32 microcontroller and the sensors, from overvoltage and provides efficient power management for portable operation.

C. EMG Sensor**:** The Electromyography (EMG) sensor detects the electrical signals produced by muscle activity in the user's forearm or hand. These analogue signals provide valuable information about voluntary muscle movements. The raw EMG data is transmitted to the microcontroller, where it is digitized and processed for gesture recognition.

D. Touch Sensor**:** The touch sensor functions as an additional input mechanism that detects simple tap or touch-based triggers from the user. It operates using a capacitive or digital switch-based sensing method and sends a high or low digital signal to the microcontroller upon activation. In this system, the touch sensor can be used to initiate commands, confirm selections, or provide an alternative control method when muscle signals are weak. Its fast response time and low power consumption make it suitable for wearable and portable applications.

E. STM32 Microcontroller: The STM32 microcontroller acts as the core processing unit of the system. It performs analogue-to-digital conversion (ADC) for the EMG signals and receives digital motion data from the IMU. The microcontroller filters and processes the input data using embedded algorithms to classify the gestures. Its low power consumption, high processing speed, and multiple peripheral interfaces make it well-suited for real-time signal processing in this application.

G. OLED Display: The OLED display provides simple visual feedback such as system status, detected signals, or message output. It features low power consumption, high contrast, and easy interface capability through I²C, making it suitable for compact and portable devices.

B. Flow Chart

The overall workflow of the proposed communication system is illustrated in the flowchart. The process begins with the EMG and touch sensors placed on the user's forearm to capture muscle activation patterns. These signals are fed into the microcontroller, where the analogue values are digitized through the ADC module. The system then analyses the timing and duration of each activation to classify them as either a dot or a dash, forming the basic Morse elements.

Once a complete sequence is detected, the Morse code is mapped into its corresponding word based on predefined combinations. To ensure secure transmission, the generated word is encrypted using the AES-128 encryption algorithm implemented directly on the microcontroller. The encrypted output is then transmitted wirelessly via the HC-05 Bluetooth module.

On the receiving end, the mobile application establishes a Bluetooth connection and retrieves the encrypted data. The Flutter-based app decrypts the message using the shared AES key and displays the recovered text to the user. This structure

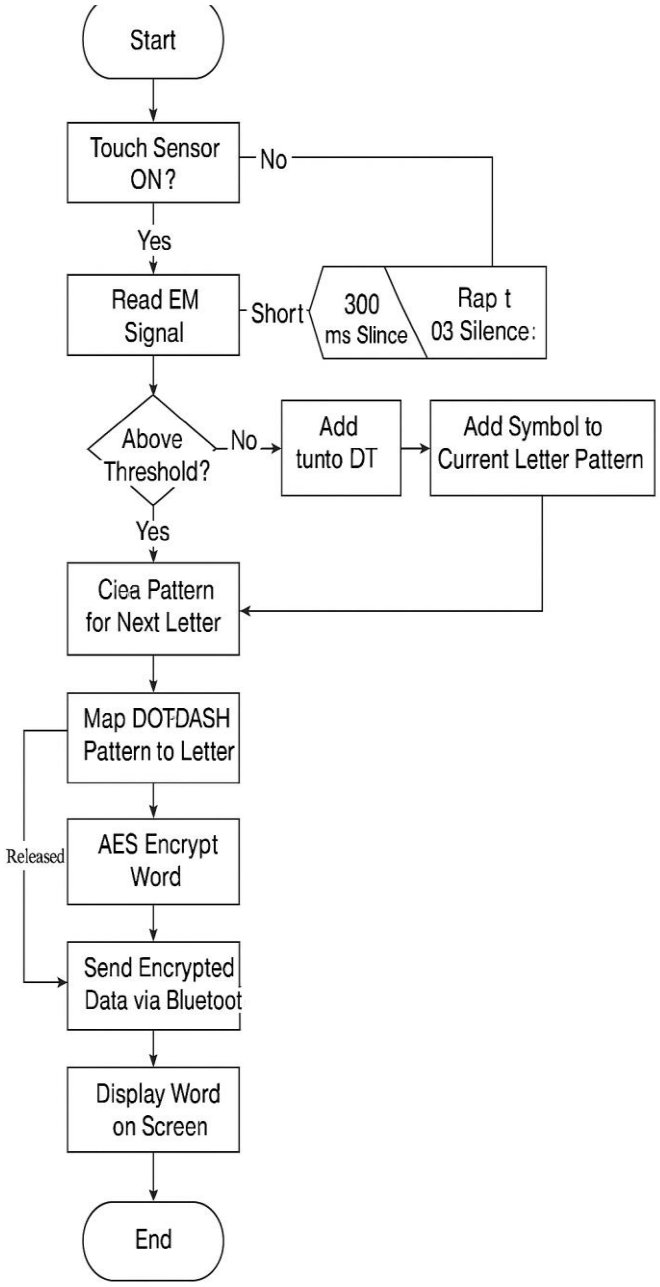

Fig 2.2 Flowchart of the developed device

enables a seamless, secure, and low-cost pipeline for translating muscle gestures into emergency words.

III. RESULTS

The following section presents the observed results, analysis of various test cases, and a summary of key findings from the EMG signal acquisition and processing stages.

A. EMG Burst Detection and Dot Dash Classification

The EMG signal recorded from the forearm was monitored for variations in muscle activity. Whenever the measured amplitude crossed a predefined threshold, the system interpreted the activation duration as either a dot or a dash. Table I summarizes the Morse symbols used in the system.

Symbol	Description
Dot (.)	Burst duration ≥ 150 ms and < 400 ms
Dash (-)	Burst duration ≥ 400 ms
Letter gap	Silence ≥ 300 ms
Word gap	Touch sensor detecting touch

Table 1: mapping morse code with dot and dashes

Short activations were classified as dots, while longer activations were classified as dashes. The system reliably differentiated between short and long muscle activations.

Fig.3.1 Morse code mapping using muscle contraction

The above figure shows morse code dot and dashes being displayed on OLED screen obtained through short duration muscle contractions showing dots(.) and longer duration muscle contractions showing dashes (-).

B. Mapping Letters

In this test case, individual letters were generated using controlled forearm muscle activations. Each letter was represented by a specific Morse code pattern, formed through combinations of short (dot) and long (dash) bursts of muscle activity. Table II summarizes the set of letters evaluated in this stage.

A	.-	J	.---	S	...
B	-...	K	-.-	T	-
C	-.-.	L	.-..	U	..-
D	-..	M	--	V	...-
E	.	N	-.	W	.--
F	..-.	O	---	X	-..-
G	--.	P	.--.	Y	-.--
H	....	Q	--.-	Z	--..
I	..	R	.-.		

Table 2: mapping morse code to letters

The results confirm that intentional muscle contractions consistently produced distinguishable activation durations, enabling accurate mapping of Morse sequences to their corresponding alphabetic characters.

Fig 3.2 : Mapped to letter 'A'

Fig 3.3 Mapped to letter 'D'

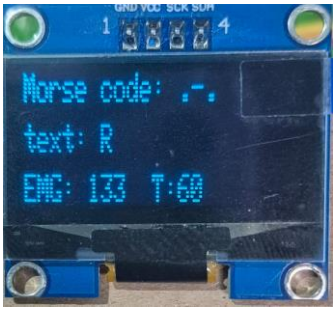

Fig 3.4 : Mapped to letter 'R'

The above figures show morse code from muscle signals being mapped to letter 'A', 'D', 'R' respectively based on the mapping table

C. Word Detection and Encrypted Output

Fig.3.5 Encrypted output for 'SOS' Fig.3.6 Encrypted output for 'HELP'

In this case, the touch sensor was used to confirm the completion of a Morse sequence and convert it into a predefined word. When the user activated the touch sensor, the system processed the accumulated Morse pattern, mapped it to the corresponding word, and immediately encrypted it using AES-128-CBC. The encrypted output was then Base64-encoded and transmitted over Bluetooth.

This approach improved reliability by preventing accidental word generation due to unintentional muscle activity. The results show that the system consistently detected the intended words and securely delivered the encrypted messages to the receiving device.

D. Decryption in Flutter Application and Display

On the receiver side, an Android application developed in Flutter established a Bluetooth connection with the HC-05 module and received the encrypted messages in the format ENC:<Base64>. The app decoded the Base64 string and applied AES-128-CBC decryption using the same key and IV as the STM32.

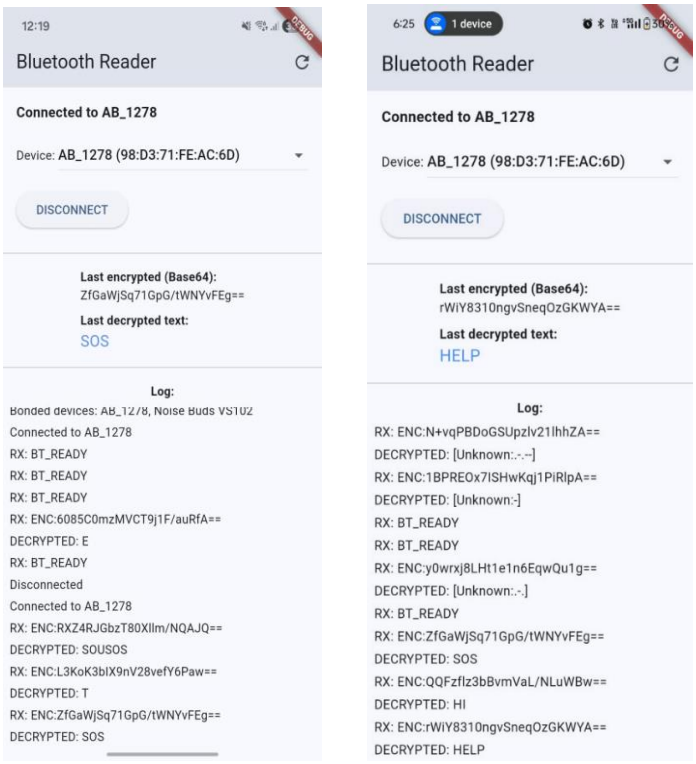

Fig. 3.7 Decryption in Flutter Application and Display for 'SOS' and 'HELP'

After decryption, the recovered plaintext was displayed on the mobile screen in real time. The results confirmed that the Flutter application reliably decoded all valid messages and provided a clear and user-friendly interface for viewing the decrypted text

IV. CONCLUSION

In this work, a muscle-based communication system was successfully developed by integrating EMG signal detection, Morse-code translation, AES encryption, and wireless message decoding through a Flutter mobile application. The EMG sensor reliably captured voluntary muscle activations, which were converted into dot dash patterns and mapped to predefined words. The system ensured secure message transmission by applying AES-128-CBC encryption before sending the data to a Bluetooth-connected mobile device. On the receiver side, the Flutter application accurately decrypted the messages and displayed the intended text in real time.

The results demonstrate that the proposed approach offers a low-cost, portable, and secure alternative communication method for individuals with speech or mobility limitations.

V. FUTURE WORK

Although the proposed EMG–Morse communication system demonstrates reliable operation, several enhancements can further improve its functionality. First, expanding the Morse-to-word mapping to support a larger vocabulary or full sentence construction would make the system more versatile for real-world communication. Incorporating adaptive thresholding or machine-learning-based signal classification could increase accuracy across different users and varying muscle strengths.

On the security side, AES encryption may be extended to include dynamic keys or mutual authentication between the microcontroller and the mobile application. The Flutter application can also be upgraded to display QR codes, maintain a message history, or integrate cloud-based storage for remote monitoring. Finally, miniaturizing the hardware and adopting a wearable form factor would improve usability and enable long-term, continuous operation in daily activities.

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